

# ARNAV GUPTA

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## PROFILE

AI/ML Developer specializing in deep learning, transformer architectures, and high-accuracy predictive systems. Experienced in building production-oriented models for fraud detection, clinical decision support, and audio classification with a strong focus on performance, reliability, and scalability. Multi-discipline achiever — Multiple Olympiad Medalist (STEM), National-Level Gold Medalist in Skating, and Black Belt in Karate — bringing the same precision and drive to AI research and real-world deployment.

## PROJECTS

### SecureBank – Fintech Fraud Detection System | Mar 2025 – Jun 2025

- Designed an uncertainty-aware fraud detection pipeline using TabNet, ensemble models, and feature engineering.
- Achieved high predictive reliability on financial datasets through model fusion and uncertainty calibration.
- Optimized inference pipeline for CPU efficiency and scalable deployment scenarios.

### SafePay AI – Real-Time Payment Fraud Prevention System | Hackathon Project

- Built a real-time AI-powered fraud detection system combining NLP-based SMS scam analysis (TF-IDF + Logistic Regression) with transaction risk scoring (Random Forest).
- Engineered a dual-model pipeline achieving ~85–90% accuracy, with explainable risk assessments providing block/approve recommendations per transaction.
- Developed full-stack application using Flask REST API and a modern dark-mode frontend with real-time risk visualization.

### ClinPath AI – Dynamic Clinical Pathway Generator | AI Clinical Decision Support (Hackathon Project)

- Engineered a rule-based clinical pathway engine integrated with a Groq-hosted LLaMA 3.1 LLM to dynamically generate patient-specific care workflows with plain-language reasoning explanations.
- Designed a safety-first, non-diagnostic architecture separating evidence-based rule logic from LLM explanation, ensuring clinical reliability.
- Built a full-stack application (Flask + React/Vite) featuring interactive flowchart visualization of generated care pathways.

### Deep Temporal Attention Network for Music Genre Classification (GTZAN) | Sep 2025 – Jan 2026

- Built a deep CNN-Transformer model leveraging temporal attention on spectrogram data.
- Achieved ~95% classification accuracy using spectrogram augmentation and attention-based modeling.
- Conducted empirical experimentation on the GTZAN dataset with optimized training workflows.

## SKILLS

**Programming:** Python, C++   **Frameworks & Tools:** PyTorch, Scikit-learn, TabNet, Flask, React   **Other:** Data Structures & Algorithms  
**ML / DL:** Machine Learning, Deep Learning, Transformers, CNNs, LLMs, Model Optimization, AI Research, NLP  
**Languages:** English (Fluent) • Marathi (Fluent) • Hindi (Fluent) • Japanese (Conversational)

## PUBLICATIONS

### Adaptive TabNet- $\beta$ -VAE Fusion Ensemble for Uncertainty-Aware Fraud Detection

Springer Lecture Notes in Networks and Systems (LNNS) – Published, ICAIN 2025

## EDUCATION

### B.Tech in Computer Science (Artificial Intelligence & Machine Learning)

Vishwakarma Institute of Technology (VIT), Pune | October 2024 – Present